3 The Laplace transform

3.1 Motivation and definition

While the Fourier transform is useful for solving myriad problems, it is limited in some aspects. One of the more important limitations is the fact that we cannot calculate the Fourier transform of many functions that grow as $t \to \infty$. This makes it unreliable when we suspect solutions that grow exponentially, for example. However, there is an easy way around this. We can simply modify the definition of the Fourier transform in such a way to dampen exponential functions as $t \to \infty$, thus giving us well defined transforms. This leads us to the following definition.

Definition 3.1. Let $f : \mathbb{R}_{\geq 0} \to \mathbb{C}$ be a function that grows at most exponentially as $t \to \infty$. The *Laplace transform* of f is given by

$$F(s) = \mathcal{L}[f] = \int_0^\infty f(t)\mathrm{e}^{-st}\,\mathrm{d}t. \tag{3.1}$$

Before diving into some examples, some comments are in order.

- Since the range of integration is $[0, \infty)$, we only require f to be defined for $t \geq 0$, as opposed to the Fourier transform. This makes the Laplace transform particularly well suited to dealing with *initial value problems*, but ill suited to problems involving functions defined over all of $\mathbb{R}$.
- The variable s is generally taken to be complex, but we need to be careful about for which values of s the integral exists. Below we will make this explicit in several examples.
- The phrase 'grows at most exponentially' means that there exist $M, N \in \mathbb{R}$ such that for all t, we have

  $$|f(t)| \leq M\mathrm{e}^{Nt}. \tag{3.2}$$

  This will guarantee that the Laplace transform exists for $\mathrm{Re}(s) > N$. We will also call these functions of *exponential-type*.

Since we only care about the behaviour of a function $f(t)$ for positive t, it is useful to define the *Heaviside step function* $H(t)$ to be

$$H(t) = \begin{cases} 1, & t \geq 0, \\ 0, & t < 0. \end{cases} \tag{3.3}$$

Herein, whenever we reference a function $f(t)$ defined over all of $\mathbb{R}$, we really are talking about $f(t)H(t)$. Similarly, we can think of a function $f : \mathbb{R}_{\geq 0} \to \mathbb{C}$ as a function defined over all of $\mathbb{R}$ that vanishes for $t < 0$.

Example 3.1. Here we will list several useful Laplace transforms, noting any restrictions on s. It is a good exercise to check these yourself.

(i) $\mathcal{L}[\mathrm{e}^{\alpha t}] = \displaystyle\int_0^\infty \mathrm{e}^{(\alpha-s)t}\,\mathrm{d}t = \frac{1}{s-\alpha}, \qquad \mathrm{Re}(s) > \mathrm{Re}(\alpha).$

(ii) $\mathcal{L}[\cos t] = \displaystyle\int_0^\infty \mathrm{e}^{-st}\cos t\,\mathrm{d}t = \frac{1}{2}\int_0^\infty \mathrm{e}^{(\mathrm{i}-s)t} + \mathrm{e}^{-(\mathrm{i}+s)t}\,\mathrm{d}t = \frac{s}{s^2+1}, \quad \mathrm{Re}(s) > 0.$

(iii) $\mathcal{L}[\sin t] = \displaystyle\int_0^\infty \mathrm{e}^{-st}\sin t\,\mathrm{d}t = \frac{1}{2\mathrm{i}}\int_0^\infty \mathrm{e}^{(\mathrm{i}-s)t} - \mathrm{e}^{-(\mathrm{i}+s)t}\,\mathrm{d}t = \frac{1}{s^2+1}, \quad \mathrm{Re}(s) > 0.$

(iv) $\mathcal{L}[H(t)] = \displaystyle\int_0^\infty \mathrm{e}^{-st}\,\mathrm{d}t = \frac{1}{s}, \quad \mathrm{Re}(s) > 0.$

(v) $\mathcal{L}[t] = \displaystyle\int_0^\infty t\mathrm{e}^{-st}\,\mathrm{d}t = \frac{1}{s^2}, \quad \mathrm{Re}(s) > 0.$

(vi) $\mathcal{L}[t^n] = \displaystyle\int_0^\infty t^n\mathrm{e}^{-st}\,\mathrm{d}t = (-1)^n\frac{\mathrm{d}^n}{\mathrm{d}s^n}\left[\int_0^\infty \mathrm{e}^{-st}\,\mathrm{d}t\right] = \frac{n!}{s^{n+1}}, \quad \mathrm{Re}(s) > 0.$

Note that n is a positive integer in the final example. For non-integer values of n, we can make the substitution $q = st$ to give

$$\mathcal{L}[t^n] = \int_0^\infty \left(\frac{q^n}{s^n}\right)\mathrm{e}^{-q}\frac{\mathrm{d}q}{s} = \frac{1}{s^{n+1}}\int_0^\infty q^n\mathrm{e}^{-q}\,\mathrm{d}q \doteq \frac{\Gamma(n+1)}{s^{n+1}}, \tag{3.4}$$

where $\Gamma(n+1)$ is a generalisation of the factorial called the *Gamma function*.

Similar to how the Fourier transforms of $L^1(\mathbb{R})$ functions vanish as $\omega \to \infty$, the Laplace transforms of exponential type functions also decay appropriately. Concretely, we have the following result.

Theorem 3.1. The Laplace transform F of an exponential-type function f obeys

$$\lim_{s\to\infty} F(s) = 0. \tag{3.5}$$

Proof. Since f is of exponential-type, we have $|f(t)| \le M\mathrm{e}^{Nt}$ for all t. Hence

$$|F(s)| \le \int_0^\infty |f(t)||\mathrm{e}^{-st}|\,\mathrm{d}t = M\int_0^\infty \mathrm{e}^{Nt}\mathrm{e}^{-st}\,\mathrm{d}t, \tag{3.6}$$

where we're taking s to be real in order to take the limit. Computing this integral we see that

$$\lim_{s\to\infty}|F(s)| = \lim_{s\to\infty}\left(\frac{M}{s-N}\right) = 0. \tag{3.7}$$

□

3.2 Transformation properties

Just like the Fourier transform, we can use substitution of variables to understand the transformation rules for Laplace transform. These derivations are much the same, up to the caveat that changing the integration range to $[0,\infty)$ introduces a few technicalities we need to deal with.

First of all, the Laplace transforms of $\alpha f(t) + \beta g(t)$ and $f(at)$ are completely analogous to the Fourier transform case. The shifted case $f(t-a)$ with $a > 0$ is similar, but we have to take into account that the Laplace transform only knows about the values of $f(t)$ for $t > 0$. We should therefore multiply $f(t-a)$ by the step function $H(t-a)$ to ensure no values of $f(t)$ for $t < 0$ contribute to the integral. Doing so gives

$$\mathcal{L}[f(t-a)H(t-a)] = \int_0^\infty f(t-a)H(t-a)\mathrm{e}^{-st}\,\mathrm{d}t = \int_a^\infty f(t-a)\mathrm{e}^{-st}\,\mathrm{d}t, \tag{3.8}$$

which changing variables to $\tau = t - a$ shows

$$\mathcal{L}[f(t-a)H(t-a)] = \mathrm{e}^{-as}\int_0^\infty f(\tau)\mathrm{e}^{-s\tau}\,\mathrm{d}\tau = \mathrm{e}^{-as}F(s). \tag{3.9}$$

In other words, delaying a function by a time $a > 0$ changes the Laplace transform by a factor of e^{-as}.

Now consider the Laplace transform of the derivative $f'(t)$. We can integrate by parts to obtain

$$\mathcal{L}[f'(t)] = \int_0^\infty f'(t)\mathrm{e}^{-st}\,\mathrm{d}t = \Big[f(t)\mathrm{e}^{-st}\Big]_0^\infty + s\int_0^\infty f(t)\mathrm{e}^{-st}\,\mathrm{d}t. \tag{3.10}$$

By hypothesis, $f(t)$ grows at most exponentially, so for large enough s we know that $f(t)\mathrm{e}^{-st}$ vanishes at infinity, but we still pick up the lower limit. Hence

$$\mathcal{L}[f'(t)] = sF(s) - f(0), \tag{3.11}$$

where $f(0)$ is understood to mean $f(0^-)$ when 0 is a point of discontinuity. Assuming $f(t)$ is n-times differentiable, we can extend this

$$\begin{aligned}\mathcal{L}[f^{(n)}(t)] &= \int_0^\infty f^{(n)}(t)\mathrm{e}^{-st}\,\mathrm{d}t = \Big[f^{(n-1)}(t)\mathrm{e}^{-st}\Big]_0^\infty + s\mathcal{L}[f^{(n-1)}(t)],\\ &= -f^{(n-1)}(0) + s\left(\Big[f^{(n-2)}(t)\mathrm{e}^{-st}\Big]_0^\infty + s\mathcal{L}[f^{(n-2)}(t)]\right).\end{aligned} \tag{3.12}$$

By repeatedly integrating by parts we keep lowering the derivative of f and introduce a factor of s. Keeping track of all the boundary terms, we see

$$\begin{aligned}\mathcal{L}[f^{(n)}(t)] &= s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \ldots - sf^{(n-2)}(0) - f^{(n-1)}(0),\\ &= s^nF(s) - \sum_{k=1}^{n} s^{n-k}f^{(k-1)}(0).\end{aligned} \tag{3.13}$$

Up to the boundary terms, we see the Laplace transform also turns differentiation into multiplication. In the opposite direction, we have

$$\mathcal{L}[t^nf(t)] = (-1)^n\frac{\mathrm{d}^n}{\mathrm{d}s^n}\left(\int_0^\infty f(t)\mathrm{e}^{-st}\,\mathrm{d}t\right) = (-1)^n\frac{\mathrm{d}^nF}{\mathrm{d}s^n}. \tag{3.14}$$

function	Laplace transform
$f(t)$	$F(s)$
$\alpha f(t) + \beta g(t)$	$\alpha F(s) + \beta G(s)$
$f(at)$	$\frac{1}{a} F\left(\frac{s}{a}\right)$
$\mathrm{e}^{-bt} f(t)$	$F(s+b)$
$f(t-a)H(t-a)$	$\mathrm{e}^{-as} F(s)$
$t^n f(t)$	$(-1)^n F^{(n)}(s)$
$f'(t)$	$sF(s) - f(0)$
$f^{(n)}(t)$	$s^n F(s) - \sum_{k=1}^{n} s^{n-k} f^{(k-1)}(0)$
$\int_0^t f(u)\,\mathrm{d}u$	$s^{-1} F(s)$
$(f * g)(t)$	$F(s)G(s)$

Table 2: The transformation properties of Laplace transforms. Here $a > 0$, $b \in \mathbb{R}$ and α, β are arbitrary complex numbers

Example 3.2. Let $f(t) = \sin t/t$. By eq. (3.14), the Laplace transform is related to the Laplace transform of $\sin t$ by

$$\mathcal{L}[\sin t] = \frac{1}{s^2+1} = -\frac{\mathrm{d}F}{\mathrm{d}s}. \tag{3.15}$$

Therefore, $F(s)$ is given by the integral

$$F(s) = -\int \frac{\mathrm{d}s}{s^2+1} = -\tan^{-1}(s) + C. \tag{3.16}$$

We can eliminate the constant of integration by noting that $F(s) \to 0$ as $s \to \infty$. Combining this with $\tan^{-1}(s) \to \pi/2$, we see that $C = \pi/2$ and the Laplace transform is

$$\mathcal{L}\left[\frac{\sin t}{t}\right] = \frac{\pi}{2} - \tan^{-1}(s) \tag{3.17}$$

The convolution theorem also holds for Laplace transform, but since we are only interested in the value of functions on $[0, \infty)$, we have a nice simplification. Suppose $f(t)$ and $g(t)$ are two functions that are zero for $t < 0$. Then the convolution takes the form

$$(f * g)(t) = \int_{-\infty}^{\infty} f(u)g(t-u)\,\mathrm{d}u = \int_0^t f(u)g(t-u)\,\mathrm{d}u, \tag{3.18}$$

since $f(u)g(t-u)$ is only non-zero when both $u \geq 0$ and $t - u \geq 0$.

Theorem 3.2. If $f, g : \mathbb{R}_{\geq 0} \to \mathbb{C}$ are functions that grow at most exponentially, then

$$\mathcal{L}[(f * g)(t)] = \mathcal{L}\left[\int_0^t f(u)g(t-u)\,\mathrm{d}u\right] = F(s)G(s). \tag{3.19}$$

Proof. We proceed similarly to the Fourier transform case. First note that if both $f(t)$ and $g(t)$ grow at most exponentially, then so does $(f * g)(t)$. The integral expression for the Laplace

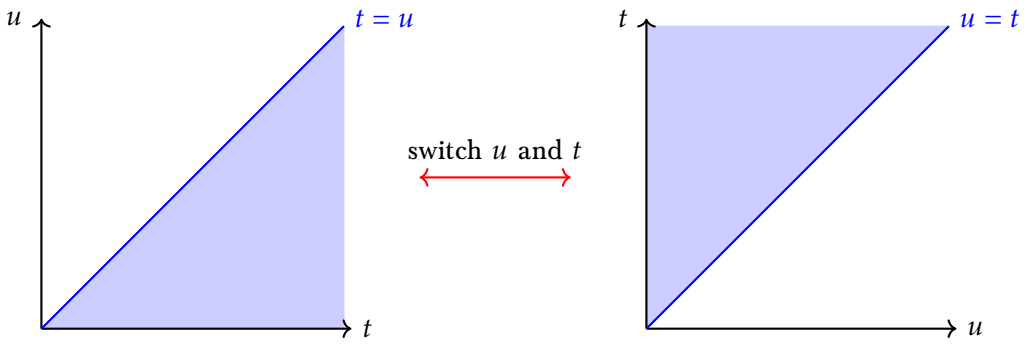


Figure 6: *Left:* The shaded area represents the domain of integration in eq. (3.20). *Right:* Changing the order of integration flips the domain so the limits change to $u \leq t < \infty$ and $0 \leq u < \infty$.

transform of $(f * g)(t)$ therefore holds. Explicitly, we have

$$\mathcal{L}[(f*g)(t)] = \int_0^\infty \left(\int_0^t f(u) g(t-u)\, \mathrm{d}u \right) \mathrm{e}^{-st}\, \mathrm{d}t. \tag{3.20}$$

We can switch the order of integration, but we have to be careful about the range of the integrals. By changing the order so we do the t integral first, we must reparametrise the domain of integration as $\{(t,u) : u \leq t < \infty, 0 \leq u < \infty\}$ (see fig. 6). Therefore, the Laplace transform is

$$\begin{aligned} \mathcal{L}[(f*g)(t)] &= \int_0^\infty \int_u^\infty f(u) g(t-u) \mathrm{e}^{-st}\, \mathrm{d}t\, \mathrm{d}u, \\ &= \int_0^\infty \int_0^\infty f(u) g(\tau) \mathrm{e}^{-s(\tau+u)}\, \mathrm{d}\tau\, \mathrm{d}u, \end{aligned} \tag{3.21}$$

where we've changed the t variable to $\tau = t - u$ in the final line. This integral factorises as

$$\mathcal{L}[(f*g)(t)] = \left(\int_0^\infty f(u) \mathrm{e}^{-su}\, \mathrm{d}u \right) \left(\int_0^\infty g(\tau) \mathrm{e}^{-s\tau}\, \mathrm{d}\tau \right) = F(s)G(s), \tag{3.22}$$

giving the result. □

Example 3.3. (i) The convolution of any function $f(t)$ with the step function $H(t)$ is simply

$$(f*H)(t) = \int_0^t f(u)\, \mathrm{d}u. \tag{3.23}$$

From the previous examples, we know that $\mathcal{L}[H(t)] = 1/s$, so the convolution theorem tells us

$$\mathcal{L}\left[\int_0^t f(u)\, \mathrm{d}u \right] = \frac{F(s)}{s}. \tag{3.24}$$

(ii) Consider the functions $f(t) = t$ and $g(t) = \mathrm{e}^t$ for $t \geq 0$. The convolution of these two

functions is given by

$$(f*g)(t) = \int_0^t u\mathrm{e}^{t-u}\,\mathrm{d}u = \mathrm{e}^t \int_0^t u\mathrm{e}^{-u}\,\mathrm{d}u = \mathrm{e}^t(-t\mathrm{e}^{-t} + 1 - \mathrm{e}^{-t}) = \mathrm{e}^t - t - 1. \tag{3.25}$$

From the previous example, we already know how to compute the Laplace transform of the right-hand side. It is given by

$$\mathcal{L}[\mathrm{e}^t - t - 1] = \frac{1}{s-1} - \frac{1}{s^2} - \frac{1}{s} = \frac{1}{(s-1)s^2} = \mathcal{L}[t]\mathcal{L}[\mathrm{e}^t], \tag{3.26}$$

matching our expectations from the convolution theorem.

Before we move on, we should note that the duality property of Fourier transforms does not generalise nicely to Laplace transforms. In particular, we have

$$\mathcal{L}^2[f(t)](u) = \int_0^\infty \int_0^\infty f(t)\mathrm{e}^{-st-us}\,\mathrm{d}t\,\mathrm{d}s = \int_0^\infty \frac{f(t)}{t+u}\,\mathrm{d}t, \tag{3.27}$$

by assuming we can change the order of integration and integrating over s. Therefore, we cannot simply read the formula sheet for Laplace transforms 'backwards' to find the Laplace transform of $F(t) = \mathcal{L}[f]$. Instead, we must (sadly) calculate it directly.

3.3 Solving ordinary differential equations

One of the main applications of the Laplace transform is using it to solve ordinary differential equations. While the Fourier transform can often be used for this purpose, the Laplace transform has some advantages. The first, which we already mentioned, is that the Laplace transform can deal with exponentially growing solutions, whereas the Fourier transform cannot. Secondly, although the boundary terms in $\mathcal{L}[y^{(n)}]$ may look cumbersome, they allow us to incorporate the initial conditions of a problem easily. This gives us better control over the solution when compared with the solution obtained from using the Fourier transform.

Example 3.4. Consider the second order ODE

$$y''(t) + y(t) = t\mathrm{e}^t, \tag{3.28}$$

subject to $y(0) = 1$ and $y'(0) = 0$. With these initial conditions, the Laplace transform of $y''(t)$ is given by $s^2Y(s) - s$, so taking the Laplace transform of the ODE gives

$$(s^2Y(s) - s) + Y(s) = \mathcal{L}[t\mathrm{e}^t] = -\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{s-1}\right) = \frac{1}{(s-1)^2}, \tag{3.29}$$

where we've used the fact $\mathcal{L}[tf(t)] = -F'(s)$ to obtain $\mathcal{L}[t\mathrm{e}^t]$. We now collect all terms involving $Y(s)$ on one side of the equation as

$$(s^2+1)Y(s) = \frac{1}{(s-1)^2} + s, \tag{3.30}$$

which then gives

$$Y(s) = \frac{1}{(s-1)^2(s^2+1)} + \frac{s}{s^2+1}. \tag{3.31}$$

All that is left is to invert this to find $y(t)$. However, while the second term is just the Laplace transform of $\cos t$, we need to do some more work to invert the first.

In order to invert $Y(s)$ in the above example, we want to rewrite it in terms of Laplace transforms we already know. Crucially, $Y(s)$ is a *rational function*, which makes this procedure fairly easy. A rational function is a function that can be written as $R(s) = P(s)/Q(s)$ where P and Q are polynomials. Assuming that $\deg(P) < \deg(Q)$, we can always perform partial fraction reduction to break up $R(s)$ into a sum of terms like $A/(s-\alpha)^n$. Let's do a simple example to recall how to do partial fractions.

Example 3.5. Consider the rational function $R(s) = s/(s-1)(s+1)^2$. First, we write out the expansion we wish to find

$$\frac{s}{(s-1)(s+1)^2} = \frac{A}{s-1} + \frac{B}{(s+1)} + \frac{C}{(s+1)^2}. \tag{3.32}$$

Now we can multiply the whole expression by $(s-1)(s+1)^2$ to eliminate the fractions

$$s = A(s+1)^2 + B(s-1)(s+1) + C(s-1). \tag{3.33}$$

There are several ways of finding A, B and C. We could expand the right-hand side and then compare coefficients to determine the unknowns. Alternatively, we can take clever values of s to obtain simpler equations and solve those instead. For example, if we take $s = 1$, all terms except for the first one vanish, so

$$(s=1): \quad 1 = 4A \Longrightarrow A = \frac{1}{4}. \tag{3.34}$$

Similarly, taking $s = -1$ makes all terms except the final one vanish, leading to

$$(s=-1): \quad -1 = -2C \Longrightarrow C = \frac{1}{2}. \tag{3.35}$$

Unfortunately, no choice of s will cause both the first and third term to vanish, so we can just take $s = 0$ to get

$$(s=0): \quad 0 = \frac{1}{4} - B - \frac{1}{2} \Longrightarrow B = -\frac{1}{4}. \tag{3.36}$$

Therefore, $R(s)$ can be written as

$$R(s) = \frac{s}{(s-1)(s+1)^2} = \frac{1}{4(s-1)} - \frac{1}{4(s+1)} + \frac{1}{2(s+1)^2}. \tag{3.37}$$

In this example, we factorised $Q(s)$ into linear factors, but this isn't actually necessary. For

example, consider the function $R(s) = 1/(s+1)(s^2+2s+2)$. The factor s^2+2s+2 has two complex roots, so instead of factorising it into linear factors we can keep it as is. Then the partial fraction decomposition takes the form

$$\frac{1}{(s+1)(s^2+2s+2)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+2s+2}. \tag{3.38}$$

The price we pay in keeping s^2+2s+2 as a quadratic is that we need to make sure to include the term Bs in the expansion. We can then proceed as usual to get

$$\frac{1}{(s+1)(s^2+2s+2)} = \frac{1}{s+1} - \frac{s+1}{s^2+2s+2}. \tag{3.39}$$

For the rest of this chapter, we will assume that $Q(s)$ can always be factorised into linear factors with *real* roots and quadratic factors with *complex* roots. With this assumption, inverting the Laplace transform is reasonably straightforward. For the linear factors, we can use

$$\mathcal{L}^{-1}\left[\frac{n!}{(s-a)^{n+1}}\right] = t^n \mathrm{e}^{at} \tag{3.40}$$

For the quadratic factors, we can always complete the square and use the inverse transforms

$$\mathcal{L}^{-1}\left[\frac{b}{(s+a)^2+b^2}\right] = \mathrm{e}^{-at}\sin bt, \quad \mathcal{L}^{-1}\left[\frac{s+a}{(s+a)^2+b^2}\right] = \mathrm{e}^{-at}\cos bt, \tag{3.41}$$

with $b \neq 0$. Both of these follow from the transformation rules we derived earlier, along with the results of example 3.1.

Example 3.4. *(continued).* Returning to the differential equation, we are now ready to invert $Y(s)$ to find the solution. We start by performing partial fractions as

$$\frac{1}{(s-1)^2(s^2+1)} = \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{Cs+D}{s^2+1}, \tag{3.42}$$

which multiplying by $(s-1)^2(s^2+1)$ gives

$$1 = A(s-1)(s^2+1) + B(s^2+1) + (Cs+D)(s-1)^2. \tag{3.43}$$

Taking $s = 1$ causes the terms with A, C and D to vanish, therefore

$$(s=1): \quad 1 = 2B \Longrightarrow B = \frac{1}{2}. \tag{3.44}$$

Since s^2+1 vanishes at $s = \mathrm{i}$, we can substitute this in to get

$$(s=\mathrm{i}): \quad 1 = (C\mathrm{i}+D)(\mathrm{i}-1)^2 = (C\mathrm{i}+D)(-2\mathrm{i}) = 2C - 2\mathrm{i}D, \tag{3.45}$$

while $s = -\mathrm{i}$ leads to

$$(s=-\mathrm{i}): \quad 1 = (-C\mathrm{i}+D)(2\mathrm{i}) = 2C + 2\mathrm{i}D. \tag{3.46}$$

Adding these two equations gives $C = 1/2$ which in turn implies $D = 0$. Finally, taking $s = 0$ then gives $A = -1/2$, hence

$$\frac{1}{(s-1)^2(s^2+1)} = -\frac{1}{2(s-1)} + \frac{1}{2(s-1)^2} + \frac{s}{2(s^2+1)}. \tag{3.47}$$

Recalling that $Y(s)$ also included an additional $s/(s^2+1)$, we get the solution $y(t)$

$$y(t) = -\frac{1}{2}e^t + \frac{1}{2}te^t + \frac{3}{2}\cos t. \tag{3.48}$$

As a sanity check, we can easily see that $y(0) = 1$ and $y'(0) = 0$.

3.4 Initial and final value theorems

Suppose that we are given a Laplace transform of a function $f(t)$ that we don't know. While it would be good to be able to find the function, it is sometimes infeasible or difficult. Despite this, it turns out that we can understand the initial value of $f(t)$ and the value of $f(t)$ at large time scales purely from the Laplace transform. This is summarised by the following two theorems.

Theorem 3.3. If $f : \mathbb{R}_{\geq 0} \to \mathbb{C}$ has Laplace transform F, then

$$\lim_{t\to 0^+} f(t) = \lim_{s\to\infty} sF(s) \tag{3.49}$$

and

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s), \tag{3.50}$$

provided the limits of $f(t)$ exist. These are the *initial value* and *final value* theorems respectively.

Proof. We start with the right-hand side and will show it equals the left. Using the integral expression for $F(s)$, we have

$$\lim_{s\to\infty} sF(s) = \lim_{s\to\infty} \int_0^\infty sf(t)e^{-st}\,dt. \tag{3.51}$$

Introducing the variable $u = st$, so $dt = du/s$, we get

$$\lim_{s\to\infty} sF(s) = \lim_{s\to\infty} \int_0^\infty f\left(\frac{u}{s}\right)e^{-u}\,du = f(0^+)\lim_{s\to\infty}\int_0^\infty e^{-u}\,du. \tag{3.52}$$

Here we've noted that $f(u/s) \to f(0^+)$ as $s \to \infty$ and pulled it out of the integral as a constant.[1] Computing the integral shows that

$$\lim_{s\to\infty} sF(s) = f(0^+) = \lim_{t\to 0^+} f(t), \tag{3.53}$$

giving the initial value theorem. A similar argument gives the final value theorem. □

[1] This step needs a little more justification in order to make it rigorous, but we will brush it under the rug.

The condition that the limit of $f(t)$ as $t \to \infty$ exists puts fairly strict conditions on the poles of $sF(s)$. These are the values of s for which $sF(s)$ is infinite. In particular, for the limit to exist, all poles of $sF(s)$ must belong to the half-plane $\mathbb{H} = \{s : \mathrm{Re}(s) < 0\}$. If this isn't the case, the limit of $sF(s)$ as $s \to 0$ may well be finite, while the limit of $f(t)$ as $t \to \infty$ won't exist.

Example 3.6. (i) Consider the function $f(t) = \sin t/t$. The limit as $t \to 0^+$ is given by

$$\lim_{t\to 0^+} \frac{\sin t}{t} = \lim_{t\to 0^+} \frac{\cos t}{1} = 1, \tag{3.54}$$

by l'Hôpital's rule. We've seen that the Laplace transform of $f(t)$ is given by $\pi/2 - \tan^{-1}(s)$, so the initial value theorem therefore states that

$$1 = \lim_{s\to\infty} s\left(\frac{\pi}{2} - \tan^{-1}(s)\right). \tag{3.55}$$

(ii) Suppose we are given the Laplace transform

$$F(s) = \frac{\pi}{s(s+\mathrm{e})(s+\sqrt{2})}. \tag{3.56}$$

Instead of trying to invert this, we can use the initial and final value theorems to learn about $f(t)$. The initial value theorem states

$$\lim_{t\to 0^+} f(t) = \lim_{s\to\infty} \frac{\pi}{(s+\mathrm{e})(s+\sqrt{2})} = 0, \tag{3.57}$$

while the final value theorem gives

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} \frac{\pi}{(s+\mathrm{e})(s+\sqrt{2})} = \frac{\pi}{\mathrm{e}\sqrt{2}}. \tag{3.58}$$

Since the only poles of $sF(s)$ are at $s = -\mathrm{e}$ and $s = -\sqrt{2}$, this will give the correct limit. As an exercise, you can invert $F(s)$ and check that the limits match.

3.5 The Dirac delta function

When modelling physical or electrical systems, we regularly need to model how the system reacts to a momentary impulse. For example, if we want to understand the motion of snooker balls colliding, we need to be able to model the moment of impact between the balls to accurately measure the change in momentum. In order to do this, it is useful to define the following tool.

Definition 3.2. The *Dirac delta function* δ (or *unit impulse*) is a generalised function characterised by the properties

$$\delta(t) = 0, \quad \forall t \neq 0 \tag{3.59}$$

and

$$\int_{-\infty}^{\infty} f(t)\delta(t)\,\mathrm{d}t = f(0), \tag{3.60}$$

for all functions $f : \mathbb{R} \to \mathbb{C}$.

As far as definitions go, this one is weird. How can a function be zero almost everywhere but somehow have an integral equal to 1? Simply put, it can't— the Dirac delta function isn't really a function and the definition above is heuristic. Mathematically, it is a ***generalised function*** or ***distribution***, both concepts that allow behaviour beyond regular functions. We won't dwell on either of these concepts, but we will show how the delta function can arise as a limit of familiar functions.

Consider the family of functions

$$D_a(t) = \frac{\chi_{[-a,a]}(t)}{2a}, \tag{3.61}$$

which has the property $\int_{-\infty}^{\infty} D_a(t)\,\mathrm{d}t = 1$ for any $a > 0$. As we take $a \to 0$, the limit of $D_a(t)$ for $t \neq 0$ is 0, while $D_a(0)$ diverges. This is precisely the behaviour we want, but we need to check this gives the correct property when integrated against a function f. For the sake of simplicity, let $f : \mathbb{R} \to \mathbb{R}$ be a continuous real-valued function, then we have

$$\min_{t\in[-a,a]} f(t) \leq \frac{1}{2a}\int_{-\infty}^{\infty} f(t)\chi_{[-a,a]}(t)\,\mathrm{d}t \leq \max_{t\in[-a,a]} f(t). \tag{3.62}$$

Since we know that f is continuous, we know that the minimum and maximum of $f(t)$ on the range $[-a, a]$ coincide under the limit $a \to 0$, meaning that both the upper and lower bound for the integral are equal when $a \to 0$. Therefore, we come to the conclusion that

$$\lim_{a\to 0}\left(\int_{-\infty}^{\infty} f(t)\frac{\chi_{[-a,a]}(t)}{2a}\,\mathrm{d}t\right) = f(0). \tag{3.63}$$

As such, we can say that the limit of $\frac{1}{2a}\chi_{[-a,a]}(t)$ is $\delta(t)$, as long as we're integrating against a function that is continuous at $t = 0$. Presenting $\delta(t)$ as a limit in this way is not unique, we have that

$$f_n(t) = \sqrt{\frac{n}{\pi}}\mathrm{e}^{-nt^2}, \quad g_n(t) = \frac{\sin nt}{\pi t}, \quad h_n(t) = \begin{cases} n(1-n|t|) & |t| < 1/n, \\ 0 & |t| \geq 1/n, \end{cases} \tag{3.64}$$

all tend to $\delta(t)$ as $n \to \infty$ in the same sense (i.e., when integrated against a function continuous at $t = 0$).

We can use the defining properties of the delta function to derive a few useful identities. First of all, we know that the integral over $\mathbb{R}$ of a delta function is equal to 1, but we also know that away from $t = 0$ the delta function is zero. Therefore, it follows that integrating over a range (a, b) will evaluate to 0 if $t = 0$ does not belong to $[a, b]$. In summary, we have

$$\int_a^b \delta(t)\,\mathrm{d}t = \begin{cases} 1, & 0 \in (a,b), \\ 0, & \text{otherwise.} \end{cases} \tag{3.65}$$

If we take the integration range to be $(-\infty, \tau]$, this implies

$$\int_{-\infty}^{\tau} \delta(t)\,\mathrm{d}t = H(\tau), \tag{3.66}$$

where $H(\tau)$ is the Heaviside step function. However, note that the integral in eq. (3.65) is ambiguous if one of the endpoints is equal to 0. Depending on the limiting function you use as a presentation for δ, you can argue for various values of the integral. To avoid this, we write

$$\int_{0^-}^{\infty} \delta(t)\,\mathrm{d}t = \lim_{h\to 0} \int_{-|h|}^{\infty} \delta(t)\,\mathrm{d}t = 1, \tag{3.67}$$

to indicate that $t = 0$ should be integrated over from below. Using this convention, the Laplace transform of $\delta(t)$ is given by

$$\mathcal{L}[\delta(t)] = \int_{0^-}^{\infty} \delta(t)\mathrm{e}^{-st}\,\mathrm{d}t = 1. \tag{3.68}$$

This is consistent with eq. (3.66). As we saw earlier, the integral of a function has Laplace transform

$$\mathcal{L}\,[H(t)] = \frac{1}{s}\mathcal{L}[\delta(t)] = \frac{1}{s}, \tag{3.69}$$

which agrees with the Laplace transform we derived for $H(t)$ earlier.

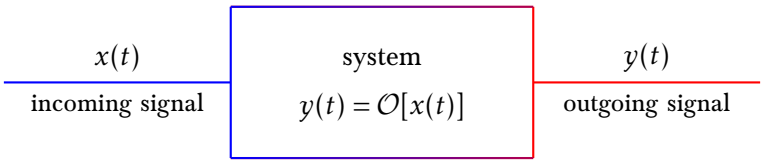
Importantly, the delta function acts as a unit for convolution. That is, we have $(f * \delta) = (\delta * f) = f$ for any (suitably well-behaved) function. We can verify this directly by noting

$$(f * \delta)(t) = \int_{-\infty}^{\infty} f(u)\delta(t-u)\,\mathrm{d}u = \int_{-\infty}^{\infty} f(t-\tau)\delta(\tau)\,\mathrm{d}\tau = f(t), \tag{3.70}$$

where we've used the substitution $\tau = t - u$. If we take $f(t)$ to be zero for $t < 0$, the same identity holds using the convention of making the lower limit 0^- we mentioned earlier.

3.6 Linear time-invariant systems

Quite often in engineering, we wish to analyse something made up of many interacting elements like a circuit. In these cases we supply an incoming signal $x(t)$ and wish to understand what the outgoing signal $y(t)$ is without understanding each interacting element separately. We call this a *system* and we think of it as an operator $\mathcal{O}$ which maps $x(t)$ to $y(t)$ and visualise it diagrammatically as below.



Right now, we're being very general: the operation $\mathcal{O}$ could send x to $x'(t)^2$, $\pi^{x(t)}$ or 0 for all we know. We therefore want to put some reasonable restrictions on the system so we can better analyse it.

Definition 3.3. A system $\mathcal{O} : x(t) \mapsto y(t)$ is *linear time-invariant (LTI)* if it satisfies the following properties.

- *Linearity.* For any two incoming signals $x_1(t)$ and $x_2(t)$ and any choice of constants $\alpha, \beta \in \mathbb{C}$, $\mathcal{O}$ satisfies

$$\mathcal{O}[\alpha x_1(t) + \beta x_2(t)] = \alpha \mathcal{O}[x_1(t)] + \beta \mathcal{O}[x_2(t)] = \alpha y_1(t) + \beta y_2(t). \tag{3.71}$$

- *Time invariance.* Delaying an incoming signal by some time $a > 0$ delays the outgoing signal by a with no other changes. Symbolically, that is

$$\mathcal{O}[x(t-a)] = y(t-a). \tag{3.72}$$

LTI systems are heavily constrained. So much so, we can actually derive a formula for $y(t)$ without specifying precisely what $\mathcal{O}$ is. To do this, we represent our incoming signal $x(t)$ as

$$x(t) = (x * \delta)(t) = \int_{0^-}^{\infty} x(\tau)\delta(t-\tau)\,\mathrm{d}\tau, \tag{3.73}$$

using the fact that the delta function is a unit for convolution. Now we apply $\mathcal{O}$ to $x(t)$, giving

$$\mathcal{O}[x(t)] = \mathcal{O}\left[\int_{0^-}^{\infty} x(\tau)\delta(t-\tau)\,\mathrm{d}\tau\right] = \int_{0^-}^{\infty} x(\tau)\mathcal{O}[\delta(t-\tau)]\,\mathrm{d}\tau, \tag{3.74}$$

where we've moved $\mathcal{O}$ into the integral because it is linear. Note that it does not act on $x(\tau)$ because it only acts on functions of t. In order to justify this step, we think of the integral as the limit of a sum and then assume $\mathcal{O}$ is suitably well-behaved so we can take this limit after introducing it. Writing $g(t) = \mathcal{O}[\delta(t)]$, we have that

$$y(t) = \int_{0^-}^{\infty} x(\tau)g(t-\tau)\,\mathrm{d}\tau, \tag{3.75}$$

where we've used the fact that the system is time invariant to write $\mathcal{O}[\delta(t-\tau)] = g(t-\tau)$. Therefore, the outgoing signal of an LTI system is given by the convolution of the incoming signal $x(t)$ with the function $g(t)$. We call $g(t)$ the *impulse response* since it measures how a system reacts to an incoming unit impulse.

When dealing with a system, it is natural to demand that the solution only depends on the present and past values of the incoming signal $x(t)$ since it wouldn't make much sense for $y(t)$ to know about future values of $x(t)$. This makes the system *causal*. However, since $y(t)$ is given as a convolution of $x(t)$ and $g(t)$, values of $x(\tau)$ for $\tau > t$ contribute to the integral, potentially violating causality. In order to avoid this, we need $g(t) = 0$ for $t < 0$, then

$$y(t) = \int_{0^-}^{t} x(\tau)g(t-\tau)\,\mathrm{d}\tau, \tag{3.76}$$

eliminating any contributions for $\tau > t$. The convolution now looks like the one we use when studying Laplace transforms, suggesting that we can use this to study LTI systems too.

We now restrict to the class of LTI systems whose outgoing signals are the solutions to a differential equation

$$a_n y^{(n)}(t) + a_{n-1} y^{(n-1)}(t) + \ldots + a_1 y'(t) + a_0 y(t) = x(t), \tag{3.77}$$

subject to the initial conditions

$$y^{(n-1)}(0) = y^{(n-2)}(0) = \ldots = y'(0) = y(0) = 0.$$

These particular initial conditions are required for the linearity of the system and indicate that the outgoing signal is totally at rest before we feed in an incoming signal. Due to these initial conditions, we have $\mathcal{L}[y^{(n)}(t)] = s^n \mathcal{L}[y(t)]$, so the Laplace transform of eq. (3.77) is

$$Y(s)(a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0) = X(s),$$

which gives the solution

$$Y(s) = \frac{X(s)}{a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0}. \tag{3.78}$$

By taking $x(t) = \delta(t)$, so $X(s) = 1$, we find that the Laplace transform of the impulse response is given by the rational function

$$G(s) = \frac{1}{a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0}, \tag{3.79}$$

giving $Y(s) = G(s)X(s)$, as one would expect from the convolution theorem. The function $G(s)$ is called the *transfer function* of the system and we will see that understanding $G(s)$ alone is enough to qualitatively understand an LTI system.

Example 3.7. Consider the LTI system whose outgoing signal is the solution to

$$y''(t) + y(t) = x(t), \tag{3.80}$$

subject to the initial conditions $y'(0) = y(0) = 0$. Laplace transforming this equation, we find the solution for $Y(s)$

$$Y(s) = \frac{X(s)}{s^2 + 1}. \tag{3.81}$$

By taking $X(s) = 1$, we see that the transfer function is $1/(s^2+1)$, from which we obtain the impulse response $g(t) = \mathcal{L}^{-1}[1/(s^2+1)] = \sin t$. Now suppose the incoming signal is given by $x(t) = t$, then the outgoing signal is given by

$$\begin{aligned} y(t) = (x * g)(t) &= \int_0^t u \sin(t-u)\,\mathrm{d}u = [u\cos(t-u)]_0^t - \int_0^t \cos(t-u)\,\mathrm{d}u, \\ &= t + [\sin(t-u)]_0^t, \\ &= t - \sin t. \end{aligned} \tag{3.82}$$

We can easily check that $y(0) = y'(0) = 0$, as expected.

3.7 Stability of LTI systems

Given an LTI system, it is often instructive to figure out the long term behaviour of the system rather than explicitly solve for various incoming signals. In particular, we want to know whether an outgoing signal grows exponentially as $t \to \infty$ or if it remains finite and is *stable*. There are various ways of phrasing stability, so we give two different definitions for LTI systems.

Definition 3.4. (i) An LTI system is *stable* if

$$\lim_{t\to\infty} g(t) = 0, \tag{3.83}$$

where $g(t)$ is the impulse response.

(ii) The system is ***bounded-input, bounded-output (BIBO)*** stable if any bounded incoming signal results in a bounded outgoing signal. That is, for any incoming $x(t)$ satisfying $|x(t)| \leq A$ for all t, there exists $B \geq 0$ such that $|y(t)| \leq B$ for all t.

In many texts, BIBO stability is reformulated as requiring

$$\|g\|_1 = \int_0^\infty |g(t)|\,\mathrm{d}t < \infty, \tag{3.84}$$

meaning $g \in L^1(\mathbb{R}_{\geq 0})$. This is equivalent to our definition, since assuming $\|g\|_1 < \infty$ gives

$$|y(t)| = |(g * x)(t)| \leq \int_0^t |g(\tau)||x(t-\tau)|\,\mathrm{d}\tau \leq A\int_0^t |g(\tau)|\,\mathrm{d}\tau, \tag{3.85}$$

where we're using $|x(t)| \leq A$ since we assume bounded input. Taking $g \in L^1(\mathbb{R}_{\geq 0})$ ensures that the right-hand side is less than or equal to $A\|g\|_1$, thus bounding the output.

For general LTI systems, BIBO stability and regular stability are different notions. The first notion of stability only requires that the system's response to a unit impulse eventually dies out, but it doesn't say anything for more general incoming signals. To see this, consider the LTI system with impulse response $1/(t+1)$, then $g(t) \to 0$ as $t \to \infty$ so the system is stable in the first sense. However, if we supply the bounded signal $x(t) = 1$ the solution is

$$y(t) = (g * x)(t) = \int_0^t \frac{1}{u+1}\,\mathrm{d}u = \log(1+t), \tag{3.86}$$

which is unbounded and shows that the system is not BIBO stable. Thankfully, for causal LTI systems with rational transfer functions the two notions are equivalent and we have the following result.

Theorem 3.4. A causal LTI system with rational transfer function $G(s) = P(s)/Q(s)$ is stable if and only if the poles of $G(s)$ belong to the half-plane $\mathbb{H} = \{s : \mathrm{Re}(s) < 0\}$.

Recall that the poles of $G(s)$ are the values of s for which $G(s) = \infty$. If $P(s)$ and $Q(s)$ do not share any roots, then these are just the values of s for which $Q(s) = 0$. The proof of this theorem is non-examinable, but the general idea is to use partial fractions to write $G(s)$ as Laplace transforms of functions $t^n e^{\alpha t}$ and then justify when $g(t)$ tends to zero.

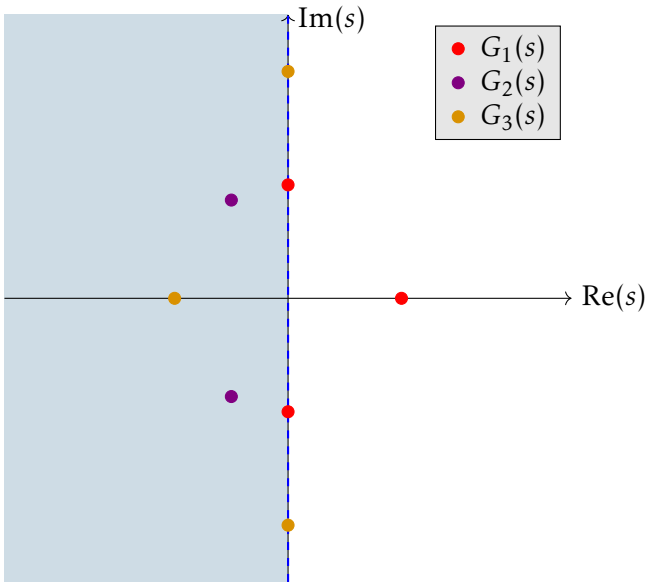


Figure 7: The poles of the transfer functions in example 3.8. The stable region $\mathrm{Re}(s) < 0$ is shaded and the boundary highlighted with a dashed blue line.

Example 3.8. Consider the stability of the systems with the following transfer functions

$$\text{(i)}\ \ G_1(s) = \frac{1}{(s-1)(s^2+1)}, \quad \text{(ii)}\ \ G_2(s) = \frac{s}{s^2+s+1}, \quad \text{(iii)}\ \ G_3(s) = \frac{s^2-s}{s^3+s^2+4s+4}. \tag{3.87}$$

(i) The poles of $G_1(s)$ occur when $s^2+1=0$ or $s-1=0$, so the roots are $s_\pm = \pm\mathrm{i}$ and $s=1$. These have real part 0 and 1 respectively, meaning the system is not stable.

(ii) The fact that the numerator isn't constant means that we should check if $P(s)$ and $Q(s)$ share any roots and reduce the fraction. This isn't the case here, as $Q(0)=1$, so the poles occur when

$$s^2+s+1=0 \Longrightarrow \left(s+\frac{1}{2}\right)^2+\frac{3}{4}=0, \tag{3.88}$$

giving poles $s_\pm = -\frac{1}{2} \pm \mathrm{i}\frac{\sqrt{3}}{2}$. Since $\mathrm{Re}(s_\pm) = -\frac{1}{2}$, this system is stable.

(iii) The zeroes of $P(s)$ are 0 and 1, but none of these have $Q(s)=0$ so we don't need to reduce the fraction. By inspection we see that $Q(-1)=0$, so $s_1=-1$ is a pole. This leads to the factorisation

$$s^3+s^2+4s+4=(s+1)(s^2+4), \tag{3.89}$$

from which we see that the other two poles are $s_\pm = \pm 2\mathrm{i}$. These have zero real part, so this system isn't stable either.